

EE2C2 FPGA Lab — Report

Student name	Student number
1	
2	
3	
4 (backup)	

Date	
Board ID	

Board ID hint: read it off the QR-code sticker on the board, the numbers in the second row to the left of the QR code. Answer each checkpoint from the lab handout in the matching box below. This lab is graded **pass/fail**; brief, correct answers are enough. Submit the compiled PDF on Brightspace.

Topic 1 — Combinational logic

Checkpoint 1a: reading the logic.

1. Identify the line that implements an **AND**, an **OR**, a **NOT**, an **XOR**, and a **multiplexer**.
2. `led_comb[1] = sw[0] & sw[1]` and `led_comb[7] = &sw` both use `&`, yet they build different gates. Explain the difference, and state exactly which switch settings drive each of these two outputs high.

Your answer

Checkpoint 1b: combinational behaviour and simulation.

1. In the waveform, when `sw` changes, do the outputs change on a clock edge or immediately? Why?
2. Why may a *testbench* change its inputs with `#10` delays, while *synthesizable* RTL must never use `#10` delays?

Your answer

Topic 2 — Sequential logic and FSM

Checkpoint 2. The FSM has a state register, next-state logic, and output decode. Why does the output decode read `state` (the registered value) rather than `next_state`?

Your answer

Topic 3 — Synthesis and implementation

Checkpoint 3. `ac701_lab.xdc` supplies two distinct kinds of information: *where* each port connects and *how fast* the clock runs. What would go wrong if the clock-period constraint were left out?

Your answer

Topic 4 — Static timing analysis

Checkpoint 4.

1. Record **WNS** and **TNS**. Is setup timing met?
2. Record the start point and endpoint of the critical path, and describe in one or two sentences what hardware lies between them.
3. Could a *hold* violation (negative WHS) be fixed by slowing the clock?
4. (Optional) Open the timing report for the worst setup path and read off the clock period T_{clk} (the requirement), the **data path delay** ($t_{\text{cq}} + t_{\text{comb}}$), the **clock uncertainty** t_{unc} , and the **slack**. Rearrange the setup-slack equation to solve for the setup-time requirement

$$t_{\text{su}} = T_{\text{clk}} - t_{\text{unc}} - (t_{\text{cq}} + t_{\text{comb}}) - t_{\text{slack}},$$

and report your four numbers and the resulting t_{su} .

Your answer

Topic 5 — Power analysis

Checkpoint 5. Imagine halving the clock frequency to 100 MHz. Which part of the total power would drop, and which would stay roughly the same?

Your answer

Topic 7 — PWM DAC

Checkpoint 7. Put a screenshot of the PWM_STUDENT_TEST_PASS line from your simulation log in the report.

Paste your PWM_STUDENT_TEST_PASS screenshot here